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Technical Specification of Biomass pellets

The Biomass pellets shall meet the technical specification as given in following table-1 and the Bidder shall submit declaration for the same as per format enclosed at **Annexure-I** of this document.

Table: 1 Technical Specification for Biomass Pellets

| S.No | Technical data                   | Unit    | Non Torrefied Biomass pellets specification                                                |
|------|----------------------------------|---------|--------------------------------------------------------------------------------------------|
| 1    | Base material                    | -       | Agro residue                                                                               |
| 2    | Shape and Size**                 | Mm      | cylindrical shape :<br>Diameter : Not more than 25 mm<br>Length : should not exceed 35 mm. |
| 3    | Fines% (length<3 mm) (ARB*)      | Wt %    | finer ≤ 5% [Less than 5% by weight]                                                        |
| 4    | Gross Calorific Value (GCV-ARB*) | kCal/kg | Acceptable range 2800 Kcal/Kg to 4000 kcal/Kg [both inclusive]                             |
| 5    | Moisture (ARB*)                  | Wt %    | Not more than 14%                                                                          |

\*ARB – As Received Basis

\*\*During contract period, diameter of pellets may be revised/modified based on result of test firing on mutual consent basis without any financial/cost implication to either NTPC or supplier.

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|-----|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.  | <b>Raw Material for pellet Manufacturing</b> | <p>9.1 Base material for pellet manufacturing shall be agro residue, which means the leftover portion of the agriculture produce, such as stubble/straw/stalk/husk of those agro residues which are surplus and not being used as animal fodder, such as paddy, soya, arhar, gwar, cotton, gram, jawar, bajara, moong, mustard, seasam, til, maize, sunflower, jute, coffee etc., groundnut shell, coconut shell, castor seed shell etc., pine needle, elephant grass, sarkanda, bamboo and its by product (e.g. Bamboo chips, Cuttings, Bamboo Dust etc.) and horticulture waste such as dry leaves and trimmings generated during the maintenance and pruning of trees and plants.</p> <p>9.2 Agro residue-based biomass pellets can be manufactured by single base material or mixing multiple base materials together.</p> <p>9.3 Mixing materials such as by product of wood work factory such as wood chips, saw dust, furniture waste etc., bagasse, press mud, molasses or natural additives/binder such as lignin, starch, and animal dung <b>may also be used with agro residue in limited proportion</b> to enhance material properties and same shall be explicitly mentioned by supplier in consignment details.</p> <p>9.4 Wood obtained from tree cutting shall not be treated as agro residue and shall be not to be used as base material or mixing purpose whatsoever.</p> <p>9.5 NTPC shall reserve the right to exclude any mixing material/additive/ binder or modify their proportion, if any adverse technical impact is observed during operation of the plant.</p> <p>9.6 If required NTPC representative can visit the pellet manufacturing unit to verify the type of raw material being used for pellet manufacturing. During visit, If NTPC team found that vendor is using raw material other than mentioned in point 9.1 , then the order of pellet supplier will be terminated</p> |
| 10. | <b>Price adjustment for GCV variation</b>    | <p>1. The Ex factory benchmark price has been notified by the MOP, which is Rs. 2.24 per 1000 Kcal basis(Excluding GST). The base rate shall be calculated based on GCV of 3400 Kcal/Kg. based on these, the per MT <b>FOR origin price</b> (Including GST) will be Rs 7996.8 per Metric Ton.</p> <p>2. The above price is including GST, which shall be paid as per the actual rate of GST as applicable from time to time (Presently 5%).</p> <p>3. The <b>transportation cost</b> shall be paid separately and the rates for the same a vendor has to quote in (Rs/MT) @3400 kcal/Kg excluding GST. The transportation cost shall be upto NTPC station and shall include all levies. The transportation rate with GST will be calculated in BOQ sheet that is FOR transportation cost. The FOR destination price (Total Price) for a vendor shall be the sum of above 3 (three) components.</p> <p>4. The price adjustment of GCV variation is applicable on FOR destination price (Total price).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |